

SUPPLY CHAIN: PROJECT CONTEXT

Bristol One City: Urban Delivery Chain

7-phase N-D-C analysis of Bristol's One City Plan delivery chain. 470,000 population. 2030 carbon-neutral target (scope 1/2/3). Four missions. City Leap £1bn+ 20-year joint venture. Bristol Climate Action Investment (4.2% LCB). Municipal sustainability-linked bond opportunity. TVPCI = 81.3% (Good zone). Bottleneck: Phase 5 (58.3%, Poor).

81.3%

Chain balance (Good; Phase 5 masked)

58.3%

Phase 5 balance (primary bottleneck)

10.7 yr

Male life expectancy gap (most vs least deprived)

52.0

e-axis score (financial abstraction: SLB gap)

City Scope and Context

Bristol is a city of approximately 470,000 people in South West England, governed by Bristol City Council and coordinated across sectors by the Bristol One City platform. The city declared a climate emergency in 2018, was the first UK local council to do so, and set a net-zero target for 2030 covering scope 1, 2, and 3 emissions. Bristol One City is not a council body: it is a cross-sector platform convened by the City Office, coordinating seven thematic boards and a Governance Board drawn from anchor institutions across public, private, voluntary, and academic sectors.

PARAMETER	VALUE	SOURCE
Population	~470,000	Bristol Key Facts 2025
Carbon neutral target	2030 (scope 1, 2, and 3)	One City Climate Strategy (Arup, 2020)
CO2 reduction since 2005	43% (city-wide)	Bristol City Council Q3 2023-24 Performance
Council direct emissions reduction	67% since 2015	Abundance Investment / BBC News (March 2025)
Most deprived population (IMD 2025)	53,100 people (11%) in 10% most deprived nationally	Bristol Key Facts November 2025
Male life expectancy gap	10.7 years (most vs least deprived, 2021-23)	PHOF 2025 (A02a)
City Leap total investment (20yr)	>£1 billion	Bristol City Leap website

PARAMETER	VALUE	SOURCE
City Leap 5yr IBP investment	£631 million	City Leap Initial Business Plan (April 2024)
BCAI raised (citizen investment)	£2 million (2 tranches, 750+ investors)	Abundance Investment / Room151 (2025)
Capital budget 2026-27	£1,669.3 million (approved)	Bristol City Council Budget 2026-27
One City Plan iterations	5 (2019, 2020, 2021, 2023, 2025)	Bristol One City website

Phase Map: 7-Phase Urban Delivery Chain

The Bristol One City delivery chain is modelled as a 7-phase system from political mandate through community uptake to measured outcomes. Unlike a physical commodity chain, D (Definition) represents the governance requirements, policy commitments, and binding constraints at each phase. C (Contribution) represents the actual delivery, measurable outputs, and community benefit achieved.

PHASE	NAME	D	C	BALANCE	N	STATUS	TRANSPARENCY	TRANSFORMATION
P0	Vision and Mandate	270	240	88.9%	226	GOOD	HIGH	Political commitment + consultation → 2050 vision, 4 missions, SDG mapping
P1	Governance and Coordination	250	210	84.0%	192	GOOD	MEDIUM	7 boards + Governance Board → coordinated cross-sector partner action
P2	Strategy and Program Design	265	240	90.6%	228	EXCELLENT	HIGH	Board mandates → City Leap IBP, Climate Strategy, LNRS, BCAI framework
P3	Capital Deployment	275	195	70.9%	164	FAIR	MEDIUM	£1.67bn budget → deployed capital in green infrastructure and services
P4	Infrastructure Delivery	260	220	84.6%	202	GOOD	HIGH	Deployed capital → heat networks, solar, retrofit, EV

PHASE	NAME	D	C	BALANCE	N	STATUS	TRANSPARENCY	TRANSFORMATION
P5	Community and Household Uptake	240	140	58.3%	107	BOTTLENECK	LOW	charging, LED lighting
								Installed infrastructure + programs → household behaviour change, deprived-community benefit
P6	Outcome Measurement	270	248	91.9%	238	EXCELLENT	HIGH	City-wide data → verified progress against 2030 target, QoL indicators, SDGs

Sustainability-Linked Bond Opportunity

Bristol has issued two tranches of a Local Climate Bond (LCB) through Abundance Investment (Bristol Climate Action Investment, BCAI), raising approximately £2 million at 4.2% fixed over 5 years. This is a use-of-proceeds instrument: funds are ring-fenced for specific green projects (heat pumps, solar, LED lighting). While successful as a community engagement mechanism, the LCB does not provide access to institutional capital markets and has no mechanism to link Bristol's coupon cost to its performance against the four One City missions.

A Sustainability-Linked Bond (SLB), structured under the ICMA Sustainability-Linked Bond Principles (SLBP, June 2024), offers a fundamentally different proposition. The coupon adjusts (steps up) if Bristol misses pre-agreed Sustainability Performance Targets (SPTs) at defined observation dates. Proceeds are general purpose. Institutional investors (pension funds, insurance companies, ESG-mandated funds) can participate at scale: typical UK municipal SLB sizes range from £50m to £200m. Bristol's existing monitoring architecture (City Leap contractual KPIs, 256 QoL indicators, PHOF, BACCC annual reviews) provides the data infrastructure required for credible KPI selection and annual verification.

Proposed SLB KPI Framework (indicative)

KPI AXIS	PROPOSED KPI	BASELINE	2027 SPT	2030 SPT	STEP-UP IF MISSED	VERIFICATION SOURCE
Environment (E)	City-wide CO2 reduction vs 2005 baseline	43% (2023)	55%	70%	25 bps	City Council annual carbon report; BACCC review

KPI AXIS	PROPOSED KPI	BASELINE	2027 SPT	2030 SPT	STEP-UP IF MISSED	VERIFICATION SOURCE
Environment (E)	City Leap zero-carbon generation capacity (MW)	0 MW (2022 baseline)	120 MW	180 MW	25 bps	City Leap contractual KPI audit (Ameresco/BCC)
Social (S)	Male life expectancy gap (most vs least deprived)	10.7 years (2021-23)	Below 10.0 yr	Below 9.0 yr	20 bps	PHOF 2025 (A02a); ONS data
Social (S)	Residents in 10% most deprived areas	53,100 (11%, IMD 2025)	Below 10.5%	Below 9.5%	15 bps	IMD update; QoL Survey deprivation decile data
Governance (G)	City Leap social value delivered	£30m (of £61.5m, ~2024)	£50m	£61.5m+	10 bps	City Leap annual social value report
Note: KPI selection, SPT calibration, and step-up triggers require ICMA SLBP external review (Second Party Opinion) before issuance. The TrueValue N-D-C framework provides the independent phase-resolved scoring methodology for annual KPI verification.						

Phase Intervention Worksheet

Provisional disclaimer: the following recommendations are derived analytically from N-D-C phase data and publicly available sources. They represent a structural diagnostic, not a capital commitment recommendation. Field validation, stakeholder co-design, and independent due diligence are required before any capital or policy decision is taken.

Five-Axis Failure Diagnosis

All five axes are below the 80-point threshold, indicating systemic stress across all dimensions of the Bristol One City delivery chain. The most critical signals are from In2 (temporal: the 2030 deadline is 4 years away and household uptake is near-zero) and e (financial: no institutional-scale bond instrument is yet structured).

CONSTANT	WHAT IT MEASURES	CHAIN SCORE	SYSTEM DIAGNOSIS	INTERVENTION CATEGORY
π	Operational balance: chain average	81.3 (above 80)	Good aggregate, but Phase 5 failure is masked. Addressing Phase 5 alone would lift π to ~84.	Community engagement infrastructure; household programme design

CONSTANT	WHAT IT MEASURES	CHAIN SCORE	SYSTEM DIAGNOSIS	INTERVENTION CATEGORY
ϕ	Proportional value distribution	71.0 (below 80)	Phase 5 N=107 vs chain avg 194: value is not reaching communities proportionally.	Just transition fund; deprived-ward targeting; community wealth building
$\sqrt{2}$	Structural overhead: D>>C friction	65.0 (below 80)	D-C gap of 100 at Phase 5, 80 at Phase 3. Requirements are binding but delivery lags.	Programme simplification; trusted intermediaries; place-based neighbourhood model
ln2	Value transformation: temporal gap	61.0 (below 80)	6+ years since 2019 declaration; 2030 deadline 4 years away; household uptake near-zero.	Accelerated delivery programme; net-zero neighbourhood demonstrators; pay-as-you-save
e	Financial abstraction: instruments linked to real metrics	52.0 (below 80)	BCAI at £2m; no SLB structured; City Leap JV is not a bond instrument.	Sustainability-linked bond framework; ICMA SLBP alignment; institutional investor engagement

Bottleneck Phase: P5: Community and Household Uptake (Balance 58.3%)

Phase ID: P5 | **Balance score:** 58.3% (Poor zone, below 61.8% threshold) | **D-C gap:** 100 points (240 vs 140). This is the largest D-C gap in the chain.

D-C imbalance diagnosis: D=240 is driven by binding policy requirements (10% mandatory BNG, pay-as-you-save model designed, £1.5m community energy fund established, BCAI accessible from £5). C=140 reflects only 750 BCAI retail investors from a population of 470,000; scope 3 emissions structurally untracked; LNRS not covering most development sites; life expectancy gap not improving; community energy uptake in early stage. The binding requirements (D) are real but the programme design has not yet connected with the lived reality of deprived communities. This is a structural problem of trusted intermediary absence and programme accessibility, not a resource gap alone.

INTERVENTION	DESCRIPTION	ESTIMATED D/C EFFECT
Net-Zero Neighbourhood demonstrators	Co-design 3 demonstrator projects in Hartcliffe/Withywood, Filwood, and Lawrence Hill (Bristol's three most deprived wards) following the Living Places Earth Net-Zero Neighbourhood model: c.1,000 properties, blended finance stack, pay-as-you-save, no upfront cost to residents. Target: household uptake at scale via a trusted, locally-rooted intermediary rather than a retail investment platform.	C up by approximately 25-30 points (from 140 to 165-170): direct measurable uptake in 3,000 properties would represent ~0.6% of Bristol households, lifting C proportionally. D unchanged (existing requirements already in place). Estimated by analogy with comparable UK demonstrators (Innovate UK Mission Net Zero, Bristol's own programme). Source: Living Places Earth place-based one-pager; Bristol City Leap Mission Net Zero programme.

INTERVENTION	DESCRIPTION	ESTIMATED D/C EFFECT
Scope 3 household tracking programme	Introduce a structured city-wide consumption-based emissions tracking methodology at ward level, building on the BACCC 2023 recommendation to address the scope 3 data gap. Partner with the University of Bristol Cabot Institute to develop a QoL Survey supplement on consumption emissions (transport, food, goods). This lifts both C (new measurable output) and the Phase 6 data architecture.	C up by approximately 10-15 points (from 140 to 150-155): creating a measurable output where currently "structurally untracked" is the status raises C even before behaviour change occurs. Source: BACCC 2023 gap analysis (implied); QoL Survey methodology (256 indicators, extensible). This is an enabling intervention, not a direct delivery intervention.
Deprived-ward targeted BCAI tranche	Issue a third BCAI tranche specifically targeting residents of the 30 Bristol wards in the 10% most deprived nationally, with a below-market entry rate (e.g. £1 minimum, ISA-eligible, with council match-contribution for residents in receipt of Universal Credit). This directly addresses the geographic and financial exclusion evident in the current 750-investor BCAI base.	C up by approximately 8-12 points: at £500 average per deprived-ward investor and 10,000 participants (2% of 53,100 deprived residents), this raises BCAI to £5-7m and broadens the community ownership base measurably. D may increase modestly (~5 points) as new administration and equity monitoring requirements are added. Net D-C effect: gap narrows by approximately 15-17 points. Source: analogy with Camden Municipal Bond (2015) and Abundance's BCAI data.
Sustainability-Linked Bond (SLB) issuance	Structure and issue a £50-100m SLB under ICMA SLBP with 5 KPIs spanning the four One City missions (carbon, City Leap MW capacity, life expectancy gap, deprivation rate, social value delivered). Coupon step-up of 25-50 bps if SPTs missed at 2027 and 2030 observation dates. Proceeds general purpose (supplement PWLB borrowing). TrueValue N-D-C framework provides independent phase-resolved scoring for annual KPI verification. This is the e-axis intervention.	e-axis score up from 52 to approximately 72-75: an SLB at £100m would represent 5% of Bristol's capital budget, creating institutional accountability for mission delivery. The N-D-C scoring methodology becomes the verification backbone. D at Phase 3 increases modestly (~10 points, new reporting obligations). C at Phase 3 increases by approximately 25-35 points (actual institutional capital mobilised). Phase 3 balance would improve from 70.9% to approximately 78-80%. Source: ICMA SLBP June 2024; Enel SLB precedent; GFI LCB Toolkit.

Combined effect estimate: If all four interventions are applied, Phase 5 C would increase from 140 to approximately 185-195 (net of the neighbourhood demonstrators + scope 3 tracking + BCAI deprived-ward tranche). Balance would improve from 58.3% to approximately 71-75% (entering Fair zone, approaching Good). Phase 3 would improve from 70.9% to approximately 78-80% via the SLB capital mobilisation. Chain balance would lift from 81.3% to approximately 85-87%. The Phase 5 signal on the In2 axis (temporal gap) would not be resolved by these interventions alone: the 2030 deadline creates irreducible urgency that requires parallel acceleration across all four interventions simultaneously.

Axis cross-reference: Phase 5 bottleneck is primarily signalled by $\sqrt{2}$ (structural overhead: $D \gg C$ friction), $\ln 2$ (temporal gap: 6+ years to near-zero household uptake), and ϕ (proportional value: Phase 5 $N=107$ vs chain average 194). The SLB intervention addresses the e -axis directly and the $\sqrt{2}$ axis partially. Neighbourhood demonstrators address $\sqrt{2}$ and $\ln 2$.

Secondary Bottleneck: P3: Capital Deployment (Balance 70.9%)

Phase ID: P3 | Balance score: 70.9% (Fair zone, below 80%) | D-C gap: 80 points (275 vs 195).

D-C imbalance diagnosis: $D=275$ is the highest in the chain: £1.67bn approved capital budget, PWLB borrowing constraints, City Leap 50/50 JV governance, ICMA Green Finance Framework requirements. $C=195$ reflects ~£30m of £61.5m social value target delivered, BCAI at only £2m, and scope 3 capital mobilisation essentially zero. The governance and regulatory architecture is fully in place (D very high) but the capital has not yet reached the communities and assets that would raise C proportionally.

INTERVENTION	DESCRIPTION	ESTIMATED D/C EFFECT
SLB issuance (e-axis)	As described in Phase 5 worksheet above: £50-100m institutional SLB under ICMA SLBP. Proceeds supplement PWLB borrowing for green capital expenditure aligned to City Leap and One City missions.	C up by 25-35 points (from 195 to 220-230): institutional capital mobilised at scale. Phase 3 balance would improve from 70.9% to approximately 78-82%. Source: ICMA SLBP; GFI LCB Toolkit; Room151 PWLB alternatives analysis.
City Funds impact capital deployment	Engage City Funds (represented on the Governance Board by Andy Street) to deploy matched impact investment alongside the SLB into the Net-Zero Neighbourhood demonstrators. Blended finance stack: SLB (senior) + City Funds (mezzanine) + community pay-as-you-save (junior).	C up by 10-15 points: adds a local anchor institution's capital to the blended stack, improving both scale and community accountability. D increases modestly (blended finance administration). Net effect: P3 gap narrows by approximately 25-30 points. Source: Living Places Earth blended finance model; City Funds governance remit.

Axis cross-reference: Phase 3 bottleneck is primarily signalled by e (financial abstraction: no SLB, BCAI only £2m) and $\sqrt{2}$ (structural overhead: high governance D with slow capital C).

Risk Factors and Opacity Sources

RISK FACTOR	PHASE	D OR C	STRUCTURAL CAUSE (NOT ATTRIBUTION)
Scope 3 emissions untracked	P5, P6	C (missing output)	Consumption-based emissions (transport, food, goods) require household-level data collection methodology not yet implemented. National statistical infrastructure does not provide city-level scope 3 data.

RISK FACTOR	PHASE	D OR C	STRUCTURAL CAUSE (NOT ATTRIBUTION)
2030 carbon target out of reach	P0, P5	D-C gap widening	BACCC 2023 progress review noted the 2030 net-zero (scope 1/2/3) target is structurally very challenging given the pace of household and scope 3 behaviour change. The target itself (D) is fixed; C is not accelerating fast enough to close the gap.
LNRS focus area coverage gap	P4, P5	C (limited reach)	West of England LNRS focus areas are parks, riverbanks, and railway margins, none of which are likely development sites. Mandatory BNG strategic significance uplift therefore does not apply to most Bristol development. Structural, not a policy failure: the LNRS mapping methodology excludes pre-existing habitats.
City Leap scope 3 social value gap	P3, P4	C (partial delivery)	£30m of £61.5m social value target delivered at approximately the midpoint of the 5-year term. The social value model relies partly on contracts delivered by local supply chain (£50m target) which requires a mature local SME pipeline not yet fully developed.
Inter-board integration nascent	P1	C (coordination gap)	The four One City missions were only adopted in October 2024. Cross-board integration is structurally new: the 7 boards have historically operated within their own thematic lanes. Mission-based integration requires new working patterns that take time to embed.

Data Sources

Bristol One City Plan 2025 (5th iteration): bristolonecity.com: Four missions, governance structure, 200+ stakeholders

Bristol One City Climate Strategy (Arup, February 2020): 10 delivery themes, 2030 net-zero scope 1/2/3: [PDF](#)

Bristol City Leap Initial Business Plan Summary (April 2024): £631m, contractual KPIs: [PDF](#)

Bristol Climate Action Investment (Abundance Investment, March 2025): 4.2% LCB, Green Finance Framework: abundanceinvestment.com

Bristol City Council Performance Framework 2025-2028: Tier 1/2 metrics, Power BI dashboard: bristol.gov.uk

Bristol Key Facts November 2025: population, deprivation, health: bristol.gov.uk

Quality of Life Survey 2024-25: 256 indicators, 34 wards, 10 deprivation deciles: bristol.gov.uk

Bristol Voluntary Local Review 2022: 140+ SDG indicators: [UN DESA](https://un.org/sdgs)

West of England Local Nature Recovery Strategy (WENP, November 2024): LNRS focus areas: wenp.org.uk

ICMA Sustainability-Linked Bond Principles (June 2024): 5 core components: icmagroup.org

Bristol Advisory Committee on Climate Change (BACCC): 12 recommendations (2023 progress review): bristolclimatenature.org

Living Places Earth: place-based net-zero advisory, David White (former City Leap Operations Director): livingplaces.earth

- ← **Hub Overview**
- Supply Chain Overview**
- N-D-C Dashboard**
- Return Chain**
- TrueValue Home**